

Krishnendu Sinha

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Group [Sinha Molecular Evolution Group](#)

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Scholar [Google Scholar profile](#)

Academic Profile

Molecular evolution and computational biology researcher developing reproducible methods for sequence analysis and evolutionary inference. Research connects evolutionary genomics, statistical phylogenetics, protein evolution, structural interpretation, and evidence-aware analysis of biological data.

Appointments

2026–present

Associate Professor of Zoology

Department of Zoology, Siksha Bhavana,
Visva-Bharati, Santiniketan

2015–2026

West Bengal Education Service

Jhargram Raj College, Jhargram, West
Bengal

Education

2011–2017

Ph.D. in Molecular Medicine

Bose Institute, Kolkata

2009–2011

M.Sc.

University of Calcutta, Kolkata

2006–2009

B.Sc.

Presidency College, Kolkata

Scholarly Record

Complete publication record and current citation information:

- [Google Scholar](#)
- [ORCID](#)
- [Scopus Author ID 55612440000](#)

Research Focus

- Evolutionary genomics, gene-family evolution, orthology, and duplicate-gene retention.
- Molecular adaptation, codon models, branch-site inference, and statistical identifiability.
- Multiple sequence alignment, learned residue-level scoring, and codon-aware analysis.
- Protein evolution, ancestral reconstruction, structural interpretation, and molecular function.
- Reproducible bioinformatics, sensitivity analysis, and evidence-aware omics research.

Research Software

BABAPPAlign

Progressive multiple sequence alignment with learned residue-level scoring, fixed protein-language-model embeddings, affine-gap dynamic programming, and codon-aware alignment.

BABAPPASnake

Reproducible workflow for orthogroup-centered episodic selection analysis, alternative alignment pathways, phylogenetic inference, follow-up testing, and robustness summaries.

Methods and Expertise

Python; sequence and phylogenetic analysis; orthology and gene-family analysis; codon-based models; ancestral reconstruction; structural bioinformatics; simulation; workflow design; reproducible research; scientific communication and research supervision.

Research Leadership

Principal Investigator, [Sinha Molecular Evolution Group](#), with research and training activities in molecular evolution, evolutionary genomics, statistical phylogenetics, sequence methods, and computational biology.